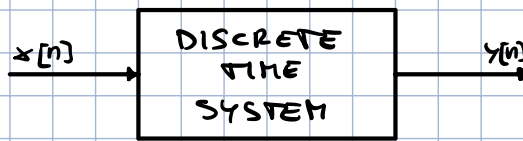


A DISCRETE TIME SYSTEM IS A COMPUTATIONAL PROCESS OR ALGORITHM THAT TRANSFORMS AN INPUT SEQUENCE $x[n]$ INTO AN OUTPUT SEQUENCE $y[n]$.

THE IMPLEMENTATION CAN BE EITHER MATHEMATICAL, HARDWARE OR SOFTWARE.



PROPERTIES

MEMORY

A SYSTEM IS MEMORY-LESS IF THE OUTPUT DEPENDS ONLY ON THE CURRENT VALUE OF THE INPUT

$$y[n] = x[n] + x[n-n_0] \quad \text{WITH MEMORY}$$

$$y[n] = x[n] + u[n-n_0] \quad \text{MEMORY-LESS}$$

$$y[n] = x[n] \alpha^{n-5} \quad \text{MEMORY-LESS}$$

$$y[n] = x[n] + x^3[n] \quad \text{MEMORY-LESS}$$

DELAY IS NOT ON INPUT!

CAUSALITY

A SYSTEM IS CAUSAL IF THE OUTPUT DEPENDS ONLY ON CURRENT AND PAST VALUES OF THE INPUT $\rightarrow$ I.E. DOES NOT LOOK INTO FUTURE OF INPUT

$$y[n] = x[n] + x[n-5] \quad \text{CAUSAL}$$

$$y[n] = x[n] + x[n+5] \quad \text{NOT CAUSAL}$$

$$y[n] = x[n] \alpha^{n-5} \quad \text{CAUSAL}$$

$$y[n] = x[n] + x^3[n] \quad \text{CAUSAL}$$

STABILITY

A SYSTEM IS STABLE IF FOR A BOUNDED INPUT CORRESPONDS A BOUNDED OUTPUT. IT IS ALSO CALLED BIBO (BOUNDED IN, BOUNDED OUT)

$$|x[n]| \leq B_x < \infty \rightarrow |y[n]| \leq B_y < \infty$$

PASSIVE/ACTIVE

A SYSTEM IS PASSIVE IF THE ENERGY OF THE OUTPUT DOES NOT EXCEED THAT OF THE INPUT

$$\sum_{n=-\infty}^{\infty} |y[n]|^2 \leq \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad E_{out} \leq E_{in}$$

CHECK IF PASSIVE OR STABLE

$$y[n] = \sum_{k=-\infty}^n x[k]$$

TO BE STABLE, THE OUTPUT MUST BE BOUNDED WHEN IT IS FED A BOUNDED INPUT.

TO CHECK INSTABILITY WE NEED TO FIND A BOUNDED SIGNAL THAT RESULTS IN AN UNBOUNDED OUTPUT:

$$x[n] = u[n] \rightarrow y[n] = \sum_{k=-\infty}^n u[k] = \sum_{k=0}^n 1 = n+1, \quad n \geq 0$$

↳ NOT STABLE, BECAUSE $|\mu[n]| \leq 1$ AND $|y[n]| \rightarrow n \rightarrow$ THERE IS NO UPPER BOUNDARY

TO BE PASSIVE, THE OUTPUT ENERGY MUST NOT EXCEED THAT OF THE INPUT:

$$x[n] = \delta[n] \rightarrow y[n] = \sum_{k=-\infty}^{\infty} \delta[k] = u[n]$$

$$E_x = \sum_{n=-\infty}^{+\infty} |\delta[n]|^2 = 1$$

$$E_y = \sum_{n=-\infty}^{+\infty} |u[n]|^2 = \infty$$

n	y[n]
0	0
1	1
2	1
	1

$$E_y > E_x \Rightarrow \text{ACTIVE, NOT PASSIVE}$$

LINEARITY

A SYSTEM IS LINEAR IF THE OUTPUT OF THE SUM OF THE INPUTS IS EQUAL TO THE SUM OF THE INDIVIDUAL OUTPUTS.

IN OTHER WORDS IT OBEYS ADDITIVITY AND SCALING.

$$\left. \begin{aligned} x_1[n] &\rightarrow y_1[n] \\ x_2[n] &\rightarrow y_2[n] \end{aligned} \right\} \quad A x_1[n] + B x_2[n] \rightarrow A y_1[n] + B y_2[n]$$

$$Y[n] = x[n] + x[n-5]$$

$$\left. \begin{aligned} Y_1[n] &= x_1[n] + x_1[n-5] \\ Y_2[n] &= x_2[n] + x_2[n-5] \end{aligned} \right\} \quad Y_{1+2}[n] = (A x_1[n] + B x_2[n]) + (A x_1[n-5] + B x_2[n-5])$$

Ex

$$Y[n] = x^2[n]$$

$$\left. \begin{array}{l} Y_1[n] = x_1^2[n] \\ Y_2[n] = x_2^2[n] \end{array} \right\} Y_{1+2}[n] = (Ax_1[n] + Bx_2[n])^2 = A^2 x_1^2[n] + B^2 x_2^2[n] + 2AB x_1[n]x_2[n]$$

$$\neq A^2 x_1^2[n] + B^2 x_2^2[n] \quad \times$$

Ex

$$Y[n] = x[n] + u[n-1]$$

$$\left. \begin{array}{l} Y_1[n] = x_1[n] + u[n-1] \\ Y_2[n] = x_2[n] + u[n-1] \end{array} \right\} Y_{1+2}[n] = (Ax_1[n] + Bx_2[n]) + u[n-1]$$

$$\neq Ax_1[n] + Au[n-1] + Bx_2[n] + Bu[n-1] \quad \times$$

Ex

$$Y[n] = x[n] + 1$$

$$\left. \begin{array}{l} x_1[n]: Y_1[n] = x_1[n] + 1 \\ x_2[n]: Y_2[n] = x_2[n] + 1 \end{array} \right\} x_1[n] + x_2[n]: Y_{1+2}[n] = (x_1[n] + x_2[n]) + 1$$

$$\neq x_1[n] + 1 + x_2[n] + 1 \quad \times$$

Ex

$$Y[n] = \alpha^{n-1} x[n]$$

$$\left. \begin{array}{l} x_1[n]: Y_1[n] = \alpha^{n-1} x_1[n] \\ x_2[n]: Y_2[n] = \alpha^{n-1} x_2[n] \end{array} \right\} Ax_1[n] + Bx_2[n]: Y_{1+2}[n] = \alpha^{n-1} (Ax_1[n] + Bx_2[n])$$

$$= A\alpha^{n-1} x_1[n] + B\alpha^{n-1} x_2[n] \quad \checkmark$$

TIME INVARIANCE

A SYSTEM IS TIME INVARIANT IF A SHIFT IN THE INPUT SIGNAL RESULTS IN AN EQUAL SHIFT OF THE OUTPUT SIGNAL.

IN OTHER WORDS IF THE I/O TRANSFORMATION DOES NOT CHANGE WITH TIME

$$x[n] \rightarrow Y[n] \Rightarrow x[n-n_0] \rightarrow Y[n-n_0]$$

Ex

$$Y[n] = x[n] + x^2[n] \quad x[n-n_0]: x[n-n_0] + x^2[n-n_0] = Y[n-n_0] \quad \checkmark$$

Ex

$$y[n] = x[n-n_0]$$

$$x[n-n_1]: x[n-n_0-n_1] = y[n-n_1] \checkmark$$

Ex

$$y[n] = x[2n]$$

$$x[n-n_0]: y_1[n] = x[2n-n_0]$$

$$y[n-n_0] = x[2(n-n_0)]$$

$$y_1[n] \neq y[n-n_0] \times$$

HERE I SHIFT THE INPUT
 $2n \rightarrow 2n-n_0$

HERE I SHIFT THE OUTPUT
 $2n \rightarrow 2(n-n_0)$

Ex

$$y[n] = x[n] + 1$$

$$x[n-n_0]: y_1[n] = x[n-n_0] + 1$$

$$y[n-n_0] = x[n-n_0] + 1$$

$$y_1[n] = y[n-n_0] \checkmark$$

Ex

$$y[n] = n x[2n]$$

$$x[n-n_0]: y_1[n] = n x[2(n-n_0)]$$

$$y[n-n_0] = (n-n_0) x[2(n-n_0)]$$

$$\neq \times$$

Ex

$$y[n] = x[n] + 5u[n-1]$$

$$x[n-n_0]: y_1[n] = x[n-n_0] + 5u[n-1]$$

$$y[n-n_0] = x[n-n_0] + 5u[n-n_0-1]$$

$$\neq \times$$

INVERTIBILITY

A SYSTEM IS INVERTIBLE IF AND ONLY IF A DISTINCT INPUT RESULTS IN A DISTINCT OUTPUT. $\rightarrow$ BIJECTIVE FUNCTION

IF A SYSTEM IS INVERTIBLE THERE EXISTS A SYSTEM CALLED "INVERSE SYSTEM" THAT PRODUCES THE OUTPUT $x[n]$ WHEN FED $y[n]$.



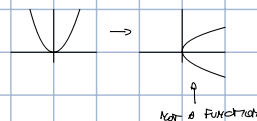
Ex

$$y[n] = x^2[n] \times$$

$$\begin{aligned} x = -a &\rightarrow y = a^2 \\ x = a &\rightarrow y = a^2 \end{aligned}$$

$$x = \pm \sqrt{y}$$

ambiguity



$$y[n] = x[n-n_0] \checkmark$$

$$y[n] = x[n] + 5u[n-1] \checkmark$$

EX

$$Y[n] = x^2[n]$$

CHECK IF:

- WITH MEMORY ✗
- CAUSAL ✓
- STABLE ✓
- PASSIVE ✗
- LINEAR ✗
- TIME INVARIANT ✓
- INVERTIBLE ✗

STABILITY:

$$|x[n]| \leq B_x < \infty \rightarrow |Y[n]| \leq B_y < \infty$$

↓
YES

PASSIVE:

$$E_y \leq E_x? \rightarrow \sum_{n=-\infty}^{+\infty} |Y[n]|^2 \leq \sum_{n=-\infty}^{+\infty} |x[n]|^2$$

$$\sum_{n=-\infty}^{+\infty} |4s^2[n]|^2 \leq \sum_{n=-\infty}^{+\infty} |2s[n]|^2 \quad \text{✗}$$

LINEAR:

$$\begin{aligned} x_1: & Y_1[n] = x_1^2[n] \\ x_2: & Y_2[n] = x_2^2[n] \\ x_1+x_2: & Y_{x_1+x_2}[n] = (x_1[n] + x_2[n])^2 = x_1^2[n] + x_2^2[n] + 2x_1[n]x_2[n] \end{aligned}$$

✗

INVERTIBLE:

$$Y[n] = x^2[n] \rightarrow x[n] = \pm \sqrt{Y[n]} \quad \text{✗}$$

↑

EX

Note: not time invariant

$$Y[n] = x[n] + 5u[n-1]$$

CHECK IF:

- WITH MEMORY ✗
- CAUSAL ✓
- STABLE ✓ BIBO
- PASSIVE ✗
- LINEAR ✗
- TIME INVARIANT ✗
- INVERTIBLE ✓

$$\begin{aligned} x_1: & Y_1[n] = x_1[n] + 5u[n-1] \\ x_2: & Y_2[n] = x_2[n] + 5u[n-1] \end{aligned} \quad \left\{ \begin{aligned} Y_1[n] + Y_2[n] &= x_1[n] + x_2[n] + 10u[n-1] \end{aligned} \right.$$

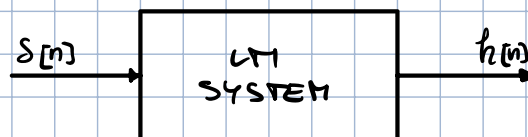
$$x_1+x_2: Y_{x_1+x_2}[n] = x_1[n] + x_2[n] + 5u[n-1] \quad \text{✗}$$

$$x[n] \rightarrow Y[n] = x[n] + 5u[n-1]$$

$$x[n-n_0] \rightarrow Y[n] = x[n-n_0] + 5u[n-1] \neq x[n-n_0] + 5u[n-n_0-1] =: Y[n-n_0]$$

LTI SYSTEMS

IF A SYSTEM IS LTI, IT CAN BE REPRESENTED BY ITS IMPULSE RESPONSE, WHICH IS THE OUTPUT OF THE SYSTEM WHEN THE INPUT IS A KRONECKER DELTA:



Ex

$$y[n] = \sum_{k=0}^{\infty} x[n-k] \longrightarrow h[n] = \sum_{k=0}^{\infty} \delta[n-k] = u[n]$$

CONVOLUTION

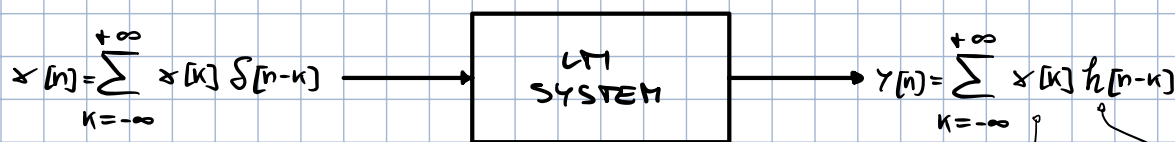
ANY SIGNAL CAN BE DECOMPOSED AND RE-WRITTEN AS A SUM OF DELTAS:

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n-k]$$

SINCE WE KNOW THE RESPONSE OF THE SYSTEM TO A DELTA AND ANY SIGNAL CAN BE DECOMPOSED INTO A SUM OF DELTAS, ALL WE NEED TO DO TO FIND THE RESPONSE OF THE SYSTEM TO ANY SIGNAL IS TO DECOMPOSE THAT SIGNAL INTO DELTAS, COMPUTE THE OUTPUT FOR EACH DELTA AND ADD ALL OUTPUTS TOGETHER.

THIS IS THE PROCESS KNOWN AS (DISCRETE TIME) CONVOLUTION.

CAN DO THIS BECAUSE SYSTEM IS LINEAR



THIS IS JUST A COEFFICIENT, THEREFORE IT DOESN'T CHANGE

THIS REACHES THE RESPONSE OF THE SYSTEM TO THE k-TH DELTA

THIS IS A VERY IMPORTANT TOOL, CALLED CONVOLUTION SUM, THAT ALLOWS US TO DIRECTLY COMPUTE THE OUTPUT OF A SYSTEM GIVEN ITS IMPULSE RESPONSE $h[n]$.

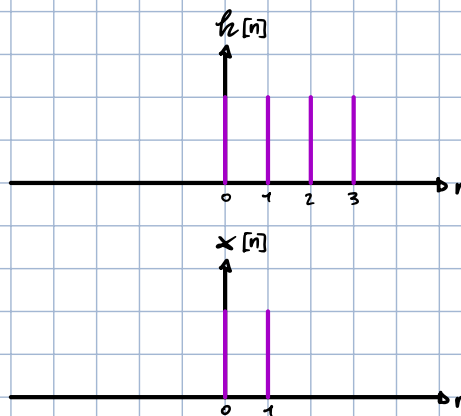
$$y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k] = x[n] * h[n]$$

Ex

$$h[n] = u[n] - u[n-4] = \sum_{k=0}^3 \delta[n-k]$$

$$x[n] = u[n] - u[n-2] = \sum_{k=0}^1 \delta[n-k]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

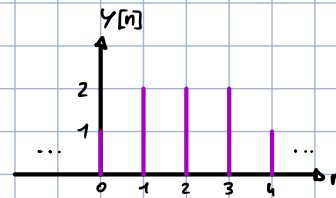


$$y[0] = \sum_{k=-\infty}^{\infty} x[k] h[-k] = x[0] h[0] = 1$$

$$y[1] = \sum_{k=-\infty}^{\infty} x[k] h[1-k] = x[0] h[1] + x[1] h[0] = 2$$

SINCE THE SIGN IS FLIPPED,
THE SIGN LOGIC OF DELAYS IS
ALSO FLIPPED $\rightarrow$ POSITIVE IS
0 DELAY TO THE RIGHT
 $[n-n_0] \rightarrow -(n-n_0) = -n+n_0$

$$y[n] = \delta[n] + 2\delta[n-1] + 2\delta[n-2] + 2\delta[n-3] + \delta[n-4]$$



$$y[2] = \sum_{k=-\infty}^{\infty} x[k] h[2-k] = x[0] h[2] + x[1] h[1] = 2$$

$$y[3] = \sum_{k=-\infty}^{\infty} x[k] h[3-k] = x[0] h[3] + x[1] h[2] = 2$$

$$y[4] = \sum_{k=-\infty}^{\infty} x[k] h[4-k] = x[0] h[4] + x[1] h[3] = 1$$

$$y[5] = 0$$

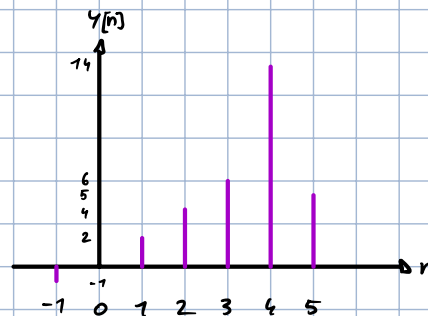
NOW SAME EXERCISE BUT IN SEQUENCE REPRESENTATION

$$x[n] = \{1 \ 2 \ 3 \ 4 \ 5\}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] y[n-k]$$

$$h[n] = \{-1 \ 2 \ 1\}$$

$$\begin{array}{l} x[n] = \quad \quad \quad 1 \ 2 \ 3 \ 4 \ 5 \\ h[n] = \quad \quad \quad \leftarrow 1 \ 2 \ 1 \rightarrow \\ \quad \quad \quad \uparrow \\ y[n] = \quad -1 \ 0 \ 2 \ 4 \ 6 \ 14 \ 5 \\ \quad \quad \quad \uparrow \\ \quad \quad \quad n=0 \end{array}$$



CONVOLUTION PROPERTIES

THE CONVOLUTION SUM OBEYS THE FOLLOWING PROPERTIES:

- COMMUTATIVE: $x_1[n] * x_2[n] = x_2[n] * x_1[n]$

- ASSOCIATIVE: $x_1[n] * x_2[n] * x_3[n] = x_1[n] * (x_2[n] * x_3[n])$

- DISTRIBUTIVE: $x_1[n] * (x_2[n] + x_3[n]) = x_1[n] * x_2[n] + x_1[n] * x_3[n]$

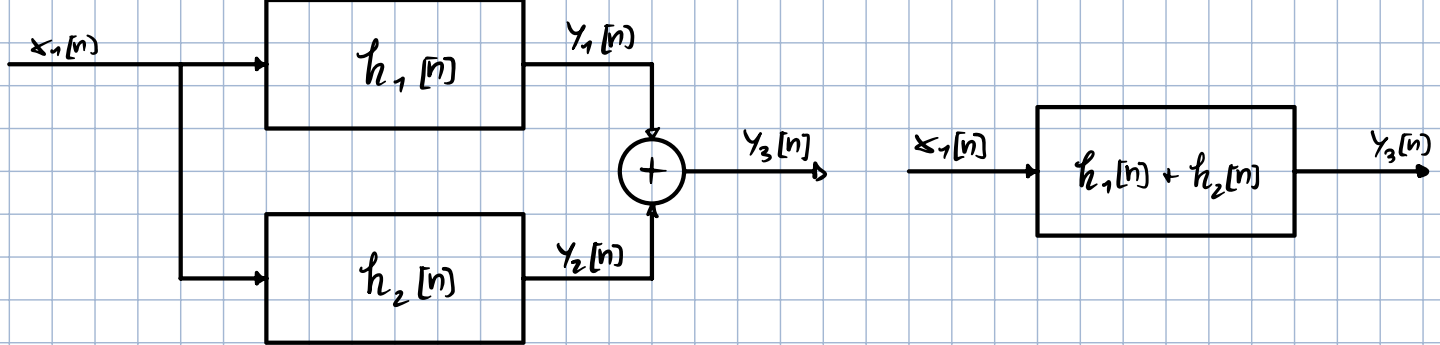
- DELAY ADDITION: $x_1[n] * x_2[n] = c[n] \longrightarrow x_1[n-N] * x_2[n-k] = c[n-N-k]$

- IDENTITY: $x[n] * \delta[n] = x[n]$

THE CONVOLUTION BETWEEN TWO DELAYED SIGNALS
IS THE CONVOLUTION OF THE ORIGINAL SIGNALS DELAYED
BY THE SUM OF THE DELAYS

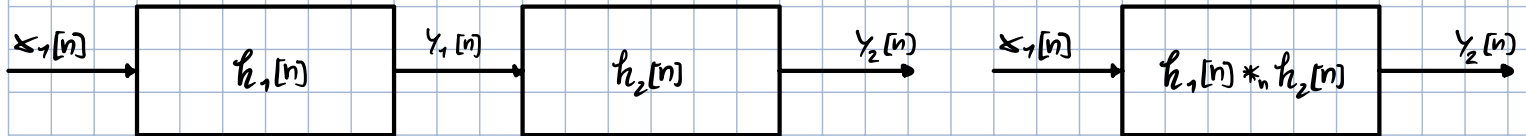
- LENGTH: IF $x_1[n]$ HAS M SAMPLES AND $x_2[n]$ HAS N SAMPLES, THEN $x_1[n] * x_2[n]$ HAS $M+N-1$ SAMPLES

SYSTEMS IN PARALLEL



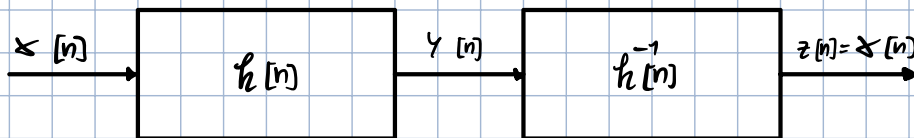
$$y_3[n] = y_1[n] + y_2[n] = x_1[n] * h_1[n] + x_1[n] * h_2[n] = x_1[n] * (h_1[n] + h_2[n])$$

SYSTEMS IN SERIES



$$y_2[n] = y_1[n] * h_2[n] = (x_1[n] * h_1[n]) * h_2[n] = x_1[n] * (h_1[n] * h_2[n])$$

INVERSE SYSTEMS



$$z[n] = y[n] * h^{-1}[n] = x[n] * h[n] * h^{-1}[n] = x[n] * \delta[n] = x[n]$$

STABILITY

AN LTI SYSTEM IS BIBO STABLE IF ITS IMPULSE RESPONSE IS ABSOLUTELY SUMMABLE:

$$|x[n]| \leq B_x < \infty \Rightarrow |y[n]| \leq B_y < \infty \iff \sum_{n=-\infty}^{+\infty} |h[n]| < \infty$$

$$|y[n]| \leq B_y < \infty$$

$$\hookrightarrow |y[n]| = |x[n] * h[n]| = \left| \sum_{k=-\infty}^{+\infty} x[k] h[n-k] \right| \leq \sum_{k=-\infty}^{+\infty} |x[k]| |h[n-k]| \leq B_x \sum_{k=-\infty}^{+\infty} |h[n-k]|$$

$$< \infty \iff \sum_{k=-\infty}^{+\infty} |h[k]| \leq B_h < \infty \quad \#$$

$$h[n] = a^n u[n]$$

IS IT STABLE?

$$\sum_{k=-\infty}^{+\infty} |h[k]| = \sum_{k=-\infty}^{+\infty} |a^k u[k]| = \sum_{k=0}^{+\infty} |a^k| = \begin{cases} \frac{1}{1-a} & |a| < 1 \rightarrow \text{STABLE} \\ \infty & |a| \geq 1 \rightarrow \text{NOT STABLE} \end{cases}$$

$$h[n] = u[n]$$

$$\sum_{k=-\infty}^{+\infty} |u[k]| = \sum_{k=0}^{\infty} 1 = \infty \quad \text{NOT STABLE}$$

$$h[n] = \frac{1}{M+N+1} \sum_{k=-M}^{+N} \delta[n-k]$$

STABLE IF ~~$M \in \mathbb{N} < \infty \cup N \in \mathbb{N} < \infty$~~

CAUSALITY

AN LTI SYSTEM IS CAUSAL IF AND ONLY IF:

$$h[n] = 0 \quad \forall n < 0$$

EX

$$y[n] = x[n-n_0] \rightarrow h[n] = \delta[n-n_0]$$

causal $\Leftrightarrow n_0 \geq 0$

$$\sum_{n=-\infty}^{+\infty} |\delta[n-n_0]| = 1 \Rightarrow \text{STABLE}$$

$$y[n] = x[n+1] - x[n] \rightarrow h[n] = \delta[n+1] - \delta[n]$$

NOT CAUSAL

$$\sum_{n=-\infty}^{+\infty} |\delta[n+1] - \delta[n]| = 2 \Rightarrow \text{STABLE}$$

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n-k] \rightarrow h[n] = \frac{1}{N} \sum_{k=0}^{N-1} \delta[n-k]$$

causal $\Leftrightarrow N \geq 1$ ($N > 0$)

$$\sum_{n=-\infty}^{+\infty} \left| \underbrace{\frac{1}{N} \sum_{k=0}^{N-1} \delta[n-k]}_{N \text{ DEGES}} \right| = \sum_{n=0}^{N-1} \left| \frac{1}{N} \sum_{k=0}^{N-1} \delta[n-k] \right| = 1 \Rightarrow \text{STABLE}$$

FIR & IIR SYSTEMS

WE CAN CHARACTERIZE SYSTEMS DEPENDING ON THEIR IMPULSE RESPONSE:

- FINITE IMPULSE RESPONSE (FIR) SYSTEMS ARE THOSE WHOSE IMPULSE RESPONSE HAS A FINITE LENGTH OF NON-ZERO VALUES
- INFINITE IMPULSE RESPONSE (IIR) SYSTEMS ARE THOSE WHOSE IMPULSE RESPONSE HAS AN INFINITE LENGTH OF NON-ZERO VALUES

IF THE SYSTEM IS CAUSAL, THE FOLLOWING SIMPLIFICATIONS CAN BE MADE:

$$\text{FIR: } y[n] = \sum_{k=-\infty}^{+\infty} h[k] x[n-k] = \sum_{k=0}^{M-1} h[k] x[n-k] \quad , \quad \text{WITH } M \text{ BEING THE LENGTH OF } h[n]$$

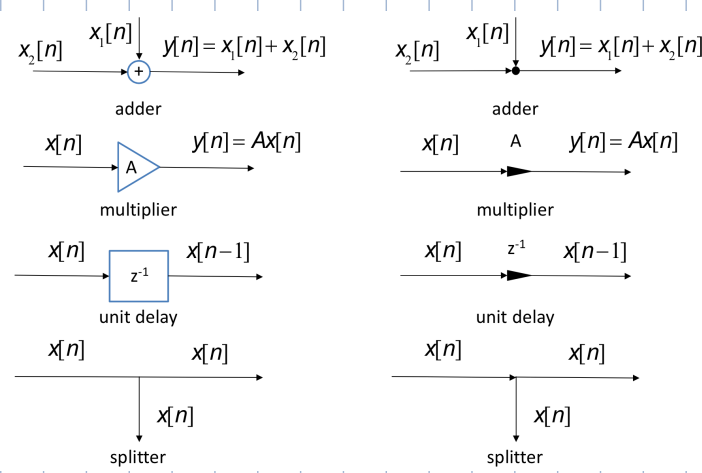
$k=-\infty$ $\curvearrowright$ $k=0$

SWAPPED TO EXPLOIT THE FACT THAT $h[n]=0 \forall n<0$ AND RESTRICT SOLUTION RANGE

NR:

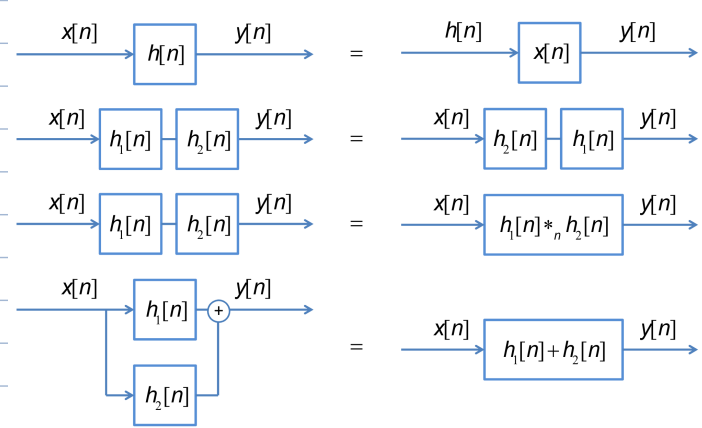
$$Y[n] = \sum_{k=-\infty}^{+\infty} h[k] x[n-k] = \sum_{k=0}^{\infty} h[k] x[n-k], \text{ LENGTH OF } h[n] \text{ IS } \infty$$

BLOCK REPRESENTATION



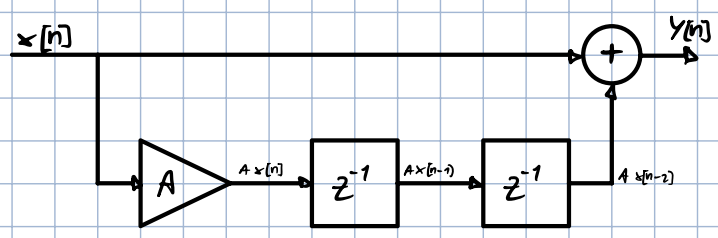
NOTE: A DELAY IS A MEMORY!

IN THE CASE OF LTI SYSTEMS, THE FOLLOWING EQUIVALENCES HOLD



EX

$$Y[n] = x[n] + A x[n-2]$$



$$Y[n] = x[n] - x[n-1] + y[n-1]$$

